

Data Story Outline: Preparing for Influenza Season

1. Beginning (why):

a. Objective:

Determine when to send staff, and how many, to each state.

b. Motivation:

The United States has an influenza season where more people than usual suffer from the flu. Some people, particularly those in vulnerable populations, develop serious complications and end up in the hospital. Hospitals and clinics need additional staff to adequately treat these extra patients. The medical staffing agency provides this temporary staff.

c. Key aspects to communicate:

- Healthcare sector taking action before next influenza season
- Vulnerable populations: serious complications ending in hospitalizations
- Hospitals in need of additional staff to treat these extra patients

d. Visualizations:

- Influenza deaths line chart (general)
- Histogram of deaths along age groups
- Pie chart of influenza deaths from vulnerable and non-vulnerable populations in the US from 2009 to 2017

2. Middle (analysis):

a. Dashboard 1: Vulnerable Population

Question: How does elderly population size impact mortality risk?

Visualization 1: Scatterplot showing correlation between population aged 65+ and total influenza deaths.

Visualization 2: Column chart of influenza related deaths in the US for vulnerable/non-vulnerable populations by year.

Insight: The risk of mortality is higher in populations 65+.

b. Dashboard 2: Geographic Distribution

Question: How is the distribution of deaths across the states?

Visualization 1: Treemap showing the Influenza Deaths of 65+ population by State and by Year.

Visualization 2: Combined map (Choropleth/symbol) showing total influenza-related deaths and 65+ Population in the United States by Year.

Insight: California, New York and Texas are the top 3 (which are also the most populated), can say also about Florida which has less deaths than expected.

c. Dashboard 3: Influenza Season

Question: When is the flu season?

Visualization: Line chart of deaths along time (months of the year)

Insight: the cold months are the ones with the highest death numbers.

Question: does the influenza season starts at the same time in every state?

Visualization: Heat Map

Insight: The states with higher deaths show a seasonal pattern, while the states with low deaths have a more stable trend around the year.

3. End (conclusions and recommendations)

a. Conclusions:

- Vulnerable population to influenza death: individuals aged 65+
- States with larger elderly population has more deaths registered.
- cold months – more deaths registered for states with higher deaths.
- States with low deaths – stable trend in year.

b. Recommendations

- More focus on states with larger elderly population
Area chart of the top 5 states with more deaths in average.
- More staff during the cold months

c. Visualization

- Bubble Chart